Indian Institute of Information Technology Bhopal

Mid-Semester Examination, March 2026

Engineering Mathematics - II

CSE/IT/ECE/MTC/PHY-2001

Semester: II

Time: 90 Min **Max. Marks: 20**

Instructions:

- All questions are compulsory.
- Mark answers number clearly.

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1. Let $\alpha \in \mathbb{R}$ be a parameter and define

$$W_\alpha = \{(x, y, z, t) \in \mathbb{R}^4 \mid x + 2y + z + t = 0,\ 2x + 4y + \alpha z + 2t = 0\}$$

and

$$A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 2 & 4 & \alpha & 2 \end{pmatrix}.$$

(a) Determine all values of α for which:

i. rows of A are linearly independent,

ii. rows of A are linearly dependent.

(b) For each case in Question (a), find a basis of W_α, and $\dim(W_\alpha)$.

(c) Determine whether there exists a value of α for which $W_\alpha = \{0\}$. Justify your answer. [5]

2. (a) Consider the system of linear equations

$$\begin{aligned} x + 2y + z &= 3, \\ 2x + 5y + 3z &= 7, \\ 3x + 7y + 4z &= 10. \end{aligned}$$

i. Determine whether the system is consistent or inconsistent.

ii. If consistent, find the general solution (using Gauss-Jordan elimination method).

(b) Let $A = \begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix}$. Find a non-singular matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix.

[2+3]

3. (a) Let

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}.$$

Consider the homogeneous system $AX = 0$.

i. Find a basis for the solution space of the system.

ii. Determine the rank and nullity of A.

(b) Let $\mathcal{P}_2(\mathbb{R})$ be the vector space of real polynomials of degree at most 2. Consider the inner product on $\mathcal{P}_2(\mathbb{R})$ by

$$\langle p, q \rangle = \int_0^1 p(x)q(x)dx.$$

Starting with the basis $\{1, x, x^2\}$, apply the Gram-Schmidt process to construct an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$ with respect to this inner product.

[2+3]

4. (a) Let A be a 3×3 matrix with eigenvalue 4 of algebraic multiplicity 2. Suppose

$$\text{rank}(A - 4I) = 2.$$

i. Find the geometric multiplicity of the eigenvalue 4.

ii. Determine whether A is diagonalizable. Justify your answer using Rank-Nullity Theorem.

(b) Let $V = \text{span}\{e^t, e^{-t}\}$ be a vector space over $\mathbb{R}$ and $D \equiv \dfrac{d}{dt}$ be a linear operator on V.

i. Find the matrix of D with respect to the basis $\{e^t, e^{-t}\}$.

ii. Does the derivative operator D have nonzero eigenvectors in V? If so, determine the eigenvalues and eigenvectors.

[2+3]